# Project due on March 5

**Popblem 1**. For the following temporal scheme,

$$y^*_{n+1} = y_n + \frac{h}{2}[3f(y_n) - f(y_{n-1})]$$

$$y_{n+1} = y_n + \frac{h}{2}[f(y_n) + f(y^*_{n+1})]$$

a) Derive the local truncation error and b) derive and plot the regions of absolute stability. Next, implement both schemes on computer. Solve the following ODEs using both schemes

$$\frac{dy_1}{dt} = -5y_1 + 3y_2$$

$$\frac{dy_2}{dt} = 100y_1 - 301y_2$$

with initial conditions $y_1(0) = 52.29$ and $y_2(0) = 83.82$. c) Is the accuracy of the solutions in the scheme, when solving the above system on computer, consistent with your analysis of local truncation error? d) Discuss your choice of $h$ when solving the above equations.

**Problem 2**. 1) Re-produce the simulation results in the Fig. 1 and Fig. 2 of the following paper: *X. Liu, S. Johnson, S. Liu, D. KanoJia, W. Yue, U. Singn, Qian Wang, Qi, Wang, Q. Nie and H. Chen. Non-linear Growth Kinetics of Breast Cancer Stem Cells: Implications for Cancer Stem Cell Targeted Therapy. 3:2473, DOI: 10.1038/srep02473, Scientific Reports, 2013.* 2) **Extra credits**: Use the model in the paper or its variation to fit (or study) one addtional set of published experimental data on tumor growth not given in this paper.